# Homework 5 Solutions

Based on Lecture 5: Angles and Trigonometric Functions.

Angles without a degree symbol are in radians. Lengths without specified units use the same units as the given lengths; areas use square units.

**Exercise 1.** A right triangle has an angle of $30^\circ$ and hypotenuse 10. Find the other two sides.

**Solution.** The side opposite $30^\circ$ is

$$10\sin 30^\circ = 10 \cdot \frac{1}{2} = \boxed{5}.$$

The other leg is

$$10\cos 30^\circ = 10 \cdot \frac{\sqrt{3}}{2} = \boxed{5\sqrt{3}}.$$

Check: $5^2 + (5\sqrt{3})^2 = 25 + 75 = 100 = 10^2$.

**Exercise 2.** An equilateral triangle has side $s$. Show that its altitude is $\dfrac{\sqrt{3}}{2}s$ and its area is $\dfrac{\sqrt{3}}{4}s^2$.

**Solution.** Let the equilateral triangle be $ABC$ and let $M$ be the midpoint of $BC$. Triangles $ABM$ and $ACM$ are congruent by SSS: $AB = AC = s$, $BM = CM = s/2$, and $AM$ is common. Thus $\angle AMB = \angle AMC$. These adjacent angles sum to $180^\circ$, so each is $90^\circ$. Consequently $AM$ is an altitude. If its length is $h$, then

$$h^2 + \left(\frac{s}{2}\right)^2 = s^2, \qquad h^2 = \frac{3}{4}s^2.$$

Because $s > 0$ and $h > 0$,

$$\boxed{h = \frac{\sqrt{3}}{2}s}, \qquad \boxed{A = \frac{1}{2}sh = \frac{\sqrt{3}}{4}s^2}.$$

**Exercise 3.** Evaluate without a calculator: $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$, $\tan^2 45^\circ - \sin^2 30^\circ$, $\dfrac{\tan 60^\circ}{\tan 30^\circ}$.

**Solution.** Substitute the special-angle values:

$$\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = \boxed{1},$$

$$\tan^2 45^\circ - \sin^2 30^\circ = 1^2 - \left(\frac{1}{2}\right)^2 = \boxed{\frac{3}{4}},$$

$$\frac{\tan 60^\circ}{\tan 30^\circ} = \frac{\sqrt{3}}{1/\sqrt{3}} = \boxed{3}.$$

**Exercise 4.** From a point 50 m from the base of a tower, the angle of elevation to the top is $60^\circ$. Find the height of the tower. If the angle were $30^\circ$, what would the height be?

**Solution.** The horizontal distance is 50 m, so $\tan\theta = h/50$. Thus

$$\boxed{h = 50\sqrt{3}\text{ m}} \quad (\theta = 60^\circ), \qquad \boxed{h = \frac{50\sqrt{3}}{3}\text{ m}} \quad (\theta = 30^\circ).$$

These are approximately 86.60 m and 28.87 m, respectively. The model measures the angle from the ground point, with no additional eye height.

**Exercise 5.** In right triangle $ABC$, $\angle C = 90^\circ$, $\angle B = 30^\circ$, and $AC = 1$. Extend side $CB$ past $B$ to a point $D$ with $BD = AB$. Show that $\angle ADC = 15^\circ$, and use triangle $ACD$ to find $\tan 15^\circ$ and $\sin 15^\circ$ exactly.

**Solution.** First $\angle A = 60^\circ$. Since $AC = 1$ is opposite $30^\circ$,

$$AB = 2, \qquad CB = \sqrt{3}, \qquad BD = AB = 2.$$

The rays $BC$ and $BD$ point in opposite directions, so $\angle ABD = 180^\circ - 30^\circ = 150^\circ$. Triangle $ABD$ is isosceles, with $AB = BD$, hence

$$\angle BAD = \angle ADB = \frac{180^\circ - 150^\circ}{2} = 15^\circ.$$

The rays $DB$ and $DC$ coincide. Therefore $\boxed{\angle ADC = 15^\circ}$. In right triangle $ACD$,

$$CD = CB + BD = \sqrt{3} + 2, \qquad AD = \sqrt{1 + (2+\sqrt{3})^2} = \sqrt{8 + 4\sqrt{3}}.$$

Consequently,

$$\boxed{\tan 15^\circ = \frac{1}{2+\sqrt{3}} = 2 - \sqrt{3}},$$

where rationalization uses $(2+\sqrt{3})(2-\sqrt{3}) = 1$. Also,

$$\sin 15^\circ = \frac{1}{\sqrt{8+4\sqrt{3}}} = \frac{1}{\sqrt{6}+\sqrt{2}} = \boxed{\frac{\sqrt{6}-\sqrt{2}}{4}}.$$

Here $(\sqrt{6}+\sqrt{2})^2 = 8 + 4\sqrt{3}$, and the positive square root is appropriate because all lengths are positive.

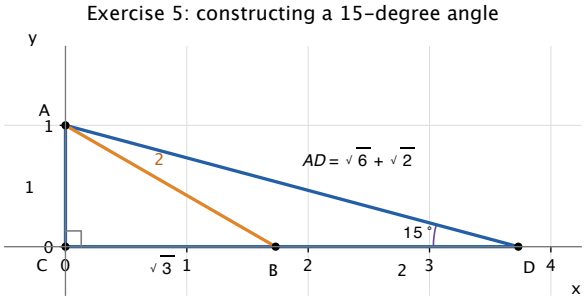



Figure 1: Right triangle ACD with A at (0,1), C at the origin, B at (root 3,0), D at (root 3 plus 2,0), AB equal to BD equal to 2, and angle ADC equal to 15 degrees

**Exercise 6.** From a point $A$ on level ground, the angle of elevation to the top of a tower is $30^\circ$. After walking 20 m straight toward the tower, the angle of elevation is $60^\circ$. Find the height of the tower.

**Solution.** Let $d$ be the distance in meters from the second observation point to the tower, and let $h$ be its height. Then

$$\frac{h}{d+20} = \tan 30^\circ = \frac{1}{\sqrt{3}}, \qquad \frac{h}{d} = \tan 60^\circ = \sqrt{3}.$$

Thus $h = (d+20)/\sqrt{3} = \sqrt{3}d$, so

$$d + 20 = 3d, \qquad d = 10, \qquad \boxed{h = 10\sqrt{3}\ \text{m}}.$$

The initial distance is 30 m; both original tangent ratios check.

**Exercise 7.** Convert $150^\circ$ and $-45^\circ$ to radians, and $\dfrac{7\pi}{6}$ and $\dfrac{5\pi}{12}$ to degrees.

**Solution.** Multiply degrees by $\pi/180$ and radians by $180/\pi$:

$$150^\circ \cdot \frac{\pi}{180^\circ} = \boxed{\frac{5\pi}{6}}, \qquad -45^\circ \cdot \frac{\pi}{180^\circ} = \boxed{-\frac{\pi}{4}},$$

$$\frac{7\pi}{6} \cdot \frac{180^\circ}{\pi} = \boxed{210^\circ}, \qquad \frac{5\pi}{12} \cdot \frac{180^\circ}{\pi} = \boxed{75^\circ}.$$

The first two answers are in radians.

**Exercise 8.** Find the angle in $[0, 2\pi)$ that is coterminal with $-\dfrac{5\pi}{3}$ and with $\dfrac{19\pi}{4}$.

**Solution.** Add or subtract full turns until the result lies in $[0, 2\pi)$:

$$-\frac{5\pi}{3} + 2\pi = \boxed{\frac{\pi}{3}}, \qquad \frac{19\pi}{4} - 4\pi = \boxed{\frac{3\pi}{4}}.$$

**Exercise 9.** A sector of a circle of radius 10 has central angle $72^\circ$. Find its arc length and area.

**Solution.** Convert first: $72^\circ = 72\pi/180 = 2\pi/5$ radians. Therefore

$$\boxed{s = r\theta = 10 \cdot \frac{2\pi}{5} = 4\pi}, \qquad \boxed{A = \frac{1}{2}r^2\theta = \frac{1}{2} \cdot 100 \cdot \frac{2\pi}{5} = 20\pi}.$$

Arc length is in the given length units; area is in their square units.

**Exercise 10.** List three angles coterminal with $135^\circ$, including one negative angle.

**Solution.** Use $135^\circ + 360^\circ k$ for integer $k$. Taking $k = -1, 1, 2$ gives

$$\boxed{-225^\circ, \quad 495^\circ, \quad 855^\circ}.$$

Each differs from $135^\circ$ by an integer multiple of $360^\circ$.

**Exercise 11.** A wheel of radius 0.4 m turns through 5 radians. How far does a point on its rim travel? Through what angle in degrees has the wheel turned?

**Solution.** The point travels along the circle through arc length

$$\boxed{s = r\theta = 0.4 \cdot 5 = 2 \text{ m}}.$$

The rotation in degrees is

$$\boxed{5 \cdot \frac{180^\circ}{\pi} = \frac{900}{\pi}^\circ \approx 286.48^\circ}.$$

This is distance traveled along the rim, rather than the straight-line displacement between the starting and ending positions.

**Exercise 12.** Find the radian measure of the angle between the hour and minute hands at 4:00, and the arc length cut off on a clock face of radius 15 cm.

**Solution.** At 4:00, the hands are four of the twelve hour divisions apart. The smaller angle is

$$\boxed{\theta = \frac{4}{12} \cdot 2\pi = \frac{2\pi}{3} \text{ radians}}.$$

The corresponding shorter arc on a circle of radius 15 cm has length

$$\boxed{s = 15 \cdot \frac{2\pi}{3} = 10\pi \text{ cm}}.$$

If the larger angle is chosen instead, it is $4\pi/3$ radians and its arc length is $20\pi$ cm.

**Exercise 13.** The minute hand turns through $2\pi$ radians every hour and the hour hand through $2\pi$ radians every 12 hours. Find each hand's angular speed in radians per minute, and find the first time after 3:00 when the two hands point in the same direction.

**Solution.** Both hands rotate clockwise. Their angular speed magnitudes are

$$\boxed{\omega_m = \frac{2\pi}{60} = \frac{\pi}{30} \text{ rad/min}}, \qquad \boxed{\omega_h = \frac{2\pi}{12 \cdot 60} = \frac{\pi}{360} \text{ rad/min}}.$$

With counterclockwise angles defined as positive, their signed angular velocities are $-\pi/30$ and $-\pi/360$ rad/min.

Let $t \geq 0$ be the number of minutes after 3:00, and measure clockwise angles from the 12 o'clock position. The minute hand starts at angle 0 and the hour hand starts at $\pi/2$. Coincidence requires

$$\frac{\pi}{30}t = \frac{\pi}{2} + \frac{\pi}{360}t + 2\pi k, \qquad k \in \mathbb{Z}.$$

Thus

$$\frac{11\pi}{360}t = \frac{\pi}{2} + 2\pi k, \qquad t = \frac{180 + 720k}{11}.$$

The least positive solution occurs at $k = 0$:

$$\boxed{t = \frac{180}{11} \text{ min} = 16\frac{4}{11} \text{ min}}.$$

The first coincidence is at $\boxed{3{:}16{:}21\frac{9}{11}}$, approximately 3:16:21.82. Indeed, the hands then have the same clockwise angle $6\pi/11$.

**Exercise 14.** A sector of a circle has perimeter 20 (two radii plus the arc). Write its area as a function of the radius $r$, and find the radius and central angle that give the largest possible area.

**Solution.** Let $s$ be the arc length and $\theta$ the central angle in radians. The perimeter condition gives

$$2r + s = 20, \qquad s = 20 - 2r.$$

For a nondegenerate sector, $r > 0$ and $s > 0$, hence $0 < r < 10$. Therefore

$$\boxed{A(r) = \frac{1}{2}rs = 10r - r^2 = 25 - (r-5)^2}.$$

Since $(r-5)^2 \geq 0$, $A(r) \leq 25$, with equality exactly when $r = 5$. At that radius $s = 20 - 2(5) = 10$, and

$$\boxed{r = 5, \qquad \theta = \frac{s}{r} = 2 \text{ radians}, \qquad A_{\max} = 25}.$$

The radius is in length units and the area in square units.

For an ordinary sector without overlapping turns, one also requires $0 < \theta \leq 2\pi$. Since $\theta = 20/r - 2$, this restricts the feasible radius to $10/(1+\pi) \leq r < 10$ (make the left endpoint strict if a full circle is excluded). The maximizing radius 5 is feasible in either convention; the maximum is unchanged.

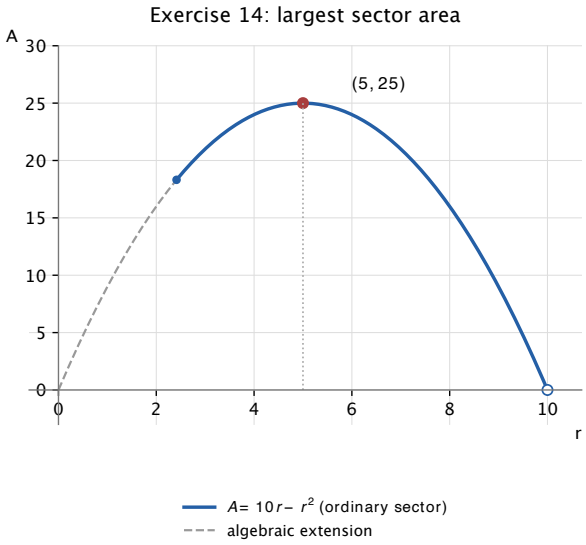



Figure 2: Sector area A equals 10r minus r squared on the feasible radii, with maximum at radius 5 and area 25; the dashed extension represents angles larger than a full turn

**Exercise 15.** The point $(5, -12)$ lies on the terminal side of $\theta$. Find all six trigonometric functions of $\theta$.

**Solution.** The distance from the origin is $r = \sqrt{5^2 + (-12)^2} = 13$. Thus

$$\boxed{\sin\theta = -\frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = -\frac{12}{5}},$$

$$\boxed{\csc\theta = -\frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = -\frac{5}{12}}.$$

All six are defined because both coordinates are nonzero. Their signs agree with quadrant IV.

**Exercise 16.** The point $\left(-\frac{8}{17}, y\right)$ lies on the unit circle in quadrant III. Find $y$ and all six trigonometric functions of the corresponding angle.

**Solution.** The unit-circle equation gives

$$\left(-\frac{8}{17}\right)^2 + y^2 = 1, \qquad y^2 = \frac{225}{289}.$$

Quadrant III requires $y < 0$, so $\boxed{y = -15/17}$. Because $r = 1$,

$$\boxed{\sin\theta = -\frac{15}{17}, \quad \cos\theta = -\frac{8}{17}, \quad \tan\theta = \frac{15}{8}},$$

$$\boxed{\csc\theta = -\frac{17}{15}, \quad \sec\theta = -\frac{17}{8}, \quad \cot\theta = \frac{8}{15}}.$$

**Exercise 17.** The terminal side of $\theta$ lies on the line $y = -2x$ in quadrant IV. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

**Solution.** Choose $(x, y) = (1, -2)$, which lies on $y = -2x$ in quadrant IV. Then $r = \sqrt{1+4} = \sqrt{5}$, and

$$\boxed{\sin\theta = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}, \quad \cos\theta = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}, \quad \tan\theta = -2}.$$

Every other point on the same quadrant-IV ray gives the same ratios.

**Exercise 18.** Evaluate, or state that the value is undefined: $\sin\pi$, $\cos\dfrac{3\pi}{2}$, $\tan\dfrac{\pi}{2}$, $\cot\pi$, $\sec 0$.

**Solution.** At $\pi$, the unit-circle point is $(-1,0)$; at $3\pi/2$ it is $(0,-1)$; at $\pi/2$ it is $(0,1)$; at $0$ it is $(1,0)$. Therefore

$$\boxed{\sin\pi = 0, \qquad \cos\frac{3\pi}{2} = 0},$$

$$\boxed{\tan\frac{\pi}{2} \text{ is undefined}, \qquad \cot\pi \text{ is undefined}, \qquad \sec 0 = 1}.$$

Here $\tan(\pi/2) = 1/0$ and $\cot\pi = (-1)/0$ have zero denominators.

**Exercise 19.** Show that the triangle with vertices $O(0,0)$, $(1,0)$, and $(\cos\theta, \sin\theta)$, where $0 < \theta < \pi$, has area $\frac{1}{2}\sin\theta$. Use this to find the area of a regular hexagon and of a regular 12-gon inscribed in the unit circle, and compare with the area of the circle.

**Solution.** Use the segment from $(0,0)$ to $(1,0)$ as the base; its length is 1. For $0 < \theta < \pi$, $\sin\theta > 0$, so the perpendicular distance from $(\cos\theta, \sin\theta)$ to the line containing the base is $\sin\theta$. This holds even if the perpendicular foot lies outside the base segment. Thus

$$\boxed{A_{\triangle} = \frac{1}{2}\cdot 1 \cdot \sin\theta = \frac{1}{2}\sin\theta}.$$

Join the center of a regular inscribed $n$-gon to its vertices. This partitions it into $n$ congruent triangles with two radii of length 1 and included angle $2\pi/n$. Rotating a triangle does not change its area, so the result just proved gives

$$A_n = \frac{n}{2}\sin\frac{2\pi}{n}.$$

Consequently,

$$\boxed{A_6 = 3\sin\frac{\pi}{3} = \frac{3\sqrt{3}}{2} \approx 2.598}, \qquad \boxed{A_{12} = 6\sin\frac{\pi}{6} = 3}.$$

The unit circle has area $\pi \approx 3.142$. Therefore

$$\frac{3\sqrt{3}}{2} < 3 < \pi.$$

Both polygons lie inside the circle, and the 12-gon gives a closer approximation to its area. The respective deficits are $\pi - 3\sqrt{3}/2 \approx 0.544$ and $\pi - 3 \approx 0.142$ square units.

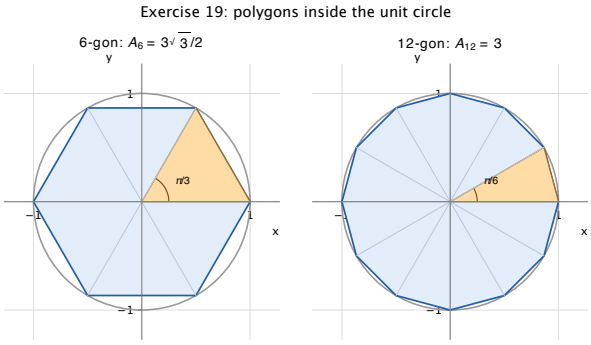



Figure 3: Regular hexagon and regular 12-gon inscribed in unit circles, with radii dividing them into triangles of central angles pi over 3 and pi over 6

**Exercise 20.** Determine the quadrant of $\theta$ if $\sin\theta < 0$ and $\tan\theta > 0$. If $\cos\theta = \dfrac{5}{13}$ and $\sin\theta < 0$, find $\sin\theta$ and $\tan\theta$.

**Solution.** For the first question, $\sin\theta < 0$ puts the terminal side in III or IV. Since $\tan\theta > 0$ requires sine and cosine to have the same sign, cosine is also negative. Thus $\boxed{\theta \text{ is in quadrant III}}$.

The second question gives a separate set of conditions. Now $\cos\theta = 5/13 > 0$ and $\sin\theta < 0$, so $\theta$ is in IV. Then

$$\sin^2\theta = 1 - \left(\frac{5}{13}\right)^2 = \frac{144}{169},$$

$$\boxed{\sin\theta = -\frac{12}{13}}, \qquad \boxed{\tan\theta = \frac{-12/13}{5/13} = -\frac{12}{5}}.$$

**Exercise 21.** Evaluate exactly: $\sin\dfrac{7\pi}{4}$, $\cos\dfrac{2\pi}{3}$, $\tan\dfrac{11\pi}{6}$, $\sin\left(-\dfrac{\pi}{3}\right)$, $\cos 210^\circ$.

**Solution.** Reduce each angle to its first-quadrant reference angle and attach the quadrant sign:

$$\sin\frac{7\pi}{4} = -\sin\frac{\pi}{4} = \boxed{-\frac{\sqrt{2}}{2}}, \qquad \cos\frac{2\pi}{3} = -\cos\frac{\pi}{3} = \boxed{-\frac{1}{2}},$$

$$\tan\frac{11\pi}{6} = -\tan\frac{\pi}{6} = \boxed{-\frac{\sqrt{3}}{3}}, \qquad \sin\left(-\frac{\pi}{3}\right) = -\sin\frac{\pi}{3} = \boxed{-\frac{\sqrt{3}}{2}},$$

$$\cos 210^\circ = -\cos 30^\circ = \boxed{-\frac{\sqrt{3}}{2}}.$$

**Exercise 22.** Evaluate exactly: $\sin\dfrac{5\pi}{4}$, $\cos\dfrac{11\pi}{6}$, $\tan\dfrac{3\pi}{4}$, $\sin 300^\circ$, $\cos\dfrac{17\pi}{3}$, $\tan\left(-\dfrac{7\pi}{6}\right)$.

**Solution.** Using symmetry and periodicity,

$$\sin\frac{5\pi}{4} = -\sin\frac{\pi}{4} = \boxed{-\frac{\sqrt{2}}{2}}, \qquad \cos\frac{11\pi}{6} = \cos\frac{\pi}{6} = \boxed{\frac{\sqrt{3}}{2}},$$

$$\tan\frac{3\pi}{4} = -\tan\frac{\pi}{4} = \boxed{-1}, \qquad \sin 300^\circ = -\sin 60^\circ = \boxed{-\frac{\sqrt{3}}{2}}.$$

Also $17\pi/3 - 4\pi = 5\pi/3$, so

$$\cos\frac{17\pi}{3} = \cos\frac{5\pi}{3} = \boxed{\frac{1}{2}}.$$

Finally, tangent has period $\pi$, hence

$$\tan\left(-\frac{7\pi}{6}\right) = \tan\left(-\frac{\pi}{6}\right) = \boxed{-\frac{\sqrt{3}}{3}}.$$

**Exercise 23.** Without a calculator, state the sign of $\sin 200^\circ$, $\cos 300^\circ$, $\tan\dfrac{5\pi}{6}$, and $\sec\left(-\dfrac{\pi}{5}\right)$.

**Solution.** The quadrant determines each sign:

- $200^\circ$ is in III, so $\boxed{\sin 200^\circ < 0}$.
- $300^\circ$ is in IV, so $\boxed{\cos 300^\circ > 0}$.
- $5\pi/6$ is in II, so $\boxed{\tan(5\pi/6) < 0}$.

- $-\pi/5$ is in IV, where cosine is positive; therefore its reciprocal satisfies $\boxed{\sec(-\pi/5) > 0}$.

**Exercise 24.** Given $\tan\theta = -\dfrac{5}{12}$ and $\sin\theta > 0$, find the other five trigonometric functions of $\theta$.

**Solution.** A negative tangent and positive sine imply negative cosine, hence quadrant II. Choose a terminal point $(-12, 5)$, so $r = 13$. The other five functions are

$$\boxed{\sin\theta = \frac{5}{13}, \quad \cos\theta = -\frac{12}{13}, \quad \csc\theta = \frac{13}{5}, \quad \sec\theta = -\frac{13}{12}, \quad \cot\theta = -\frac{12}{5}}.$$

Their quotient checks: $(5/13)/(-12/13) = -5/12$.

**Exercise 25.** Use symmetry, not a calculator, to evaluate

$$\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \cdots + \sin 359^\circ \quad \text{and} \quad \sum_{k=0}^{11} \cos\frac{k\pi}{6}.$$

**Solution.** Reflection across the $x$-axis gives $\sin(360^\circ - \alpha) = -\sin\alpha$. Pair the terms at $k^\circ$ and $(360-k)^\circ$ for $k = 1, \ldots, 179$. The only remaining term is $\sin 180^\circ = 0$. Thus

$$\sum_{k=1}^{359} \sin k^\circ = \sum_{k=1}^{179} (\sin k^\circ + \sin(360-k)^\circ) + \sin 180^\circ = \boxed{0}.$$

For the cosine sum, pair index $k$ with $k+6$ for $k = 0, \ldots, 5$. The angles differ by $\pi$, so a half-turn changes the sign of cosine:

$$\sum_{k=0}^{11} \cos\frac{k\pi}{6} = \sum_{k=0}^{5} \left(\cos\frac{k\pi}{6} + \cos\left(\frac{k\pi}{6} + \pi\right)\right) = \boxed{0}.$$

All twelve terms are used exactly once.

**Exercise 26.** Find all $\theta$ in $[0, 2\pi)$ with $2\sin^2\theta - \sin\theta - 1 = 0$.

**Solution.** Factor the equation in $u = \sin\theta$:

$$2u^2 - u - 1 = (2u+1)(u-1) = 0.$$

Thus $\sin\theta = 1$ or $\sin\theta = -1/2$. On $[0, 2\pi)$, the first condition gives $\theta = \pi/2$. The second gives the reference angle $\pi/6$ in quadrants III and IV, namely $7\pi/6$ and $11\pi/6$. Therefore all the solutions are

$$\boxed{\theta \in \left\{\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\right\}}.$$

Each sine value is a root of the factored polynomial, so all three check.

**Exercise 27.** If $\tan\theta = 2$ and $\pi < \theta < \dfrac{3\pi}{2}$, find $\sin\theta$ and $\cos\theta$.

**Solution.** In quadrant III, both coordinates are negative and tangent is positive. Choose $(x, y) = (-1, -2)$, which gives $y/x = 2$ and $r = \sqrt{5}$. Hence

$$\boxed{\sin\theta = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}, \qquad \cos\theta = -\frac{1}{\sqrt{5}} = -\frac{\sqrt{5}}{5}}.$$

Their squared values sum to $4/5 + 1/5 = 1$, and their quotient is 2.

**Exercise 28.** Simplify $\sin\theta\sec\theta\cot\theta$ and $(1+\sin\theta)(1-\sin\theta)$.

**Solution.** For the first expression, $\sec\theta$ requires $\cos\theta \neq 0$, and $\cot\theta$ requires $\sin\theta \neq 0$. On this original domain,

$$\sin\theta \sec\theta \cot\theta = \sin\theta \cdot \frac{1}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta} = \boxed{1}.$$

The cancellation does not assign a value to the original expression at excluded angles: its domain remains $\theta \neq k\pi/2$ for $k \in \mathbb{Z}$.

The second expression is defined for every real $\theta$. By difference of squares and the Pythagorean identity,

$$(1+\sin\theta)(1-\sin\theta) = 1 - \sin^2\theta = \boxed{\cos^2\theta}.$$

**Exercise 29.** Prove $\dfrac{\sec\theta - \cos\theta}{\sin\theta} = \tan\theta$ and $\dfrac{\cos\theta}{1-\sin\theta} = \dfrac{1+\sin\theta}{\cos\theta}$.

**Solution.** For the first identity, the original left side requires both $\cos\theta \neq 0$ and $\sin\theta \neq 0$. Under those conditions,

$$\frac{\sec\theta - \cos\theta}{\sin\theta} = \frac{\frac{1}{\cos\theta} - \cos\theta}{\sin\theta} = \frac{1-\cos^2\theta}{\sin\theta\cos\theta} = \frac{\sin^2\theta}{\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \boxed{\tan\theta}.$$

Thus the identity holds on the original domain $\theta \neq k\pi/2$, $k \in \mathbb{Z}$; simplification does not extend that domain.

For the second identity, the common domain of the displayed sides is $\cos\theta \neq 0$. This also guarantees $\sin\theta \neq \pm 1$, so both $1-\sin\theta$ and $1+\sin\theta$ are nonzero. Multiply the left side by $(1+\sin\theta)/(1+\sin\theta)$:

$$\frac{\cos\theta}{1-\sin\theta} = \frac{\cos\theta(1+\sin\theta)}{1-\sin^2\theta} = \frac{\cos\theta(1+\sin\theta)}{\cos^2\theta} = \boxed{\frac{1+\sin\theta}{\cos\theta}}.$$

The left side by itself is defined whenever $\sin\theta \neq 1$; in particular, it is 0 when $\sin\theta = -1$. The right side is undefined there. The identity is asserted only where both sides are defined, $\theta \neq \pi/2 + k\pi$, $k \in \mathbb{Z}$.

**Exercise 30.** Write $\cos\theta$ and $\tan\theta$ in terms of $\sin\theta = s$, assuming $\theta$ is in quadrant II.

**Solution.** Quadrant II requires $0 < s < 1$ and $\cos\theta < 0$. From $\cos^2\theta = 1 - s^2$,

$$\boxed{\cos\theta = -\sqrt{1-s^2}, \qquad \tan\theta = -\frac{s}{\sqrt{1-s^2}}}, \qquad 0 < s < 1.$$

The square root is positive on this interval, so tangent is defined and negative, as quadrant II requires.

**Exercise 31.** Use cofunction relations to show $\sin 20^\circ = \cos 70^\circ$, and find $x$ if $\sin(3x) = \cos(x + 10^\circ)$ with $0^\circ < x < 30^\circ$.

**Solution.** The cofunction identity gives

$$\sin 20^\circ = \cos(90^\circ - 20^\circ) = \boxed{\cos 70^\circ}.$$

For the equation, all angles below are in degrees. Rewrite it as

$$\sin(3x) = \sin(90^\circ - (x + 10^\circ)) = \sin(80^\circ - x).$$

Because $0^\circ < x < 30^\circ$, both $3x$ and $80^\circ - x$ lie strictly between $0^\circ$ and $90^\circ$. Sine is one-to-one on that interval: on the first-quadrant unit-circle arc, the height increases strictly as the angle increases. Therefore

$$3x = 80^\circ - x, \qquad 4x = 80^\circ, \qquad \boxed{x = 20^\circ}.$$

This lies in the required interval; substituting gives $\sin 60^\circ = \cos 30^\circ = \sqrt{3}/2$. Injectivity proves there are no other solutions in the interval.

**Exercise 32.** Use even/odd and periodicity to evaluate $\cos\left(-\dfrac{\pi}{3}\right)$, $\sin\left(\dfrac{13\pi}{6}\right)$, and $\tan\left(-\dfrac{5\pi}{4}\right)$.

**Solution.** Cosine is even, sine has period $2\pi$, and tangent is odd with period $\pi$. Thus

$$\cos\left(-\frac{\pi}{3}\right) = \cos\frac{\pi}{3} = \boxed{\frac{1}{2}},$$

$$\sin\frac{13\pi}{6} = \sin\left(2\pi + \frac{\pi}{6}\right) = \boxed{\frac{1}{2}},$$

$$\tan\left(-\frac{5\pi}{4}\right) = -\tan\frac{5\pi}{4} = -\tan\frac{\pi}{4} = \boxed{-1}.$$

**Exercise 33.** Suppose $\sin\theta + \cos\theta = \dfrac{1}{5}$ and $0 < \theta < \pi$. Find $\sin\theta\cos\theta$, decide which quadrant $\theta$ is in, and then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

**Solution.** Write $s = \sin\theta$ and $c = \cos\theta$. Squaring $s + c = 1/5$ gives

$$\frac{1}{25} = (s+c)^2 = s^2 + c^2 + 2sc = 1 + 2sc,$$

so

$$\boxed{sc = -\frac{12}{25}}.$$

Since $0 < \theta < \pi$, $s > 0$. The negative product then forces $c < 0$, so $\boxed{\theta \text{ is in quadrant II}}$.

Now

$$(s-c)^2 = s^2 + c^2 - 2sc = 1 + \frac{24}{25} = \frac{49}{25}.$$

Because $s > 0$ and $c < 0$, $s - c > 0$, giving $s - c = 7/5$. Add and subtract this equation and $s + c = 1/5$:

$$2s = \frac{8}{5}, \qquad 2c = -\frac{6}{5}.$$

Therefore

$$\boxed{\sin\theta = \frac{4}{5}, \qquad \cos\theta = -\frac{3}{5}, \qquad \tan\theta = -\frac{4}{3}}.$$

Check: $4/5 - 3/5 = 1/5$, $16/25 + 9/25 = 1$, and the signs match quadrant II.

**Exercise 34.** Reflecting a point across the line $y = x$ swaps its coordinates. Explain why this reflection sends the unit-circle point at angle $\alpha$ to the point at angle $\frac{\pi}{2} - \alpha$, and deduce the cofunction relations. Then use the quarter-turn $(x, y) \mapsto (-y, x)$ to show $\sin\left(\frac{\pi}{2} + \alpha\right) = \cos\alpha$ and $\cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha$.

**Solution.** Let $P = (\cos\alpha, \sin\alpha)$. Reflecting across $y = x$ interchanges its coordinates, producing $P' = (\sin\alpha, \cos\alpha)$. Geometrically, the reflecting line makes angle $\pi/4$ with the positive $x$-axis. Reflection reverses a ray's signed angular displacement from that line. A ray at angle $\alpha$ therefore goes to the ray at angle

$$\frac{\pi}{4} - \left(\alpha - \frac{\pi}{4}\right) = \frac{\pi}{2} - \alpha \pmod{2\pi}.$$

Thus $P'$ is also the unit-circle point $(\cos(\pi/2 - \alpha), \sin(\pi/2 - \alpha))$. Equating its coordinates gives

$$\boxed{\cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha, \qquad \sin\left(\frac{\pi}{2} - \alpha\right) = \cos\alpha}.$$

This reflection argument applies to every real $\alpha$, including angles outside quadrant I.

A counterclockwise quarter-turn adds $\pi/2$ to the angle and sends $(x, y)$ to $(-y, x)$. Hence it sends $P$ to $(-\sin\alpha, \cos\alpha)$, also equal to $(\cos(\pi/2 + \alpha), \sin(\pi/2 + \alpha))$. Therefore

$$\boxed{\sin\left(\frac{\pi}{2} + \alpha\right) = \cos\alpha, \qquad \cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha}.$$

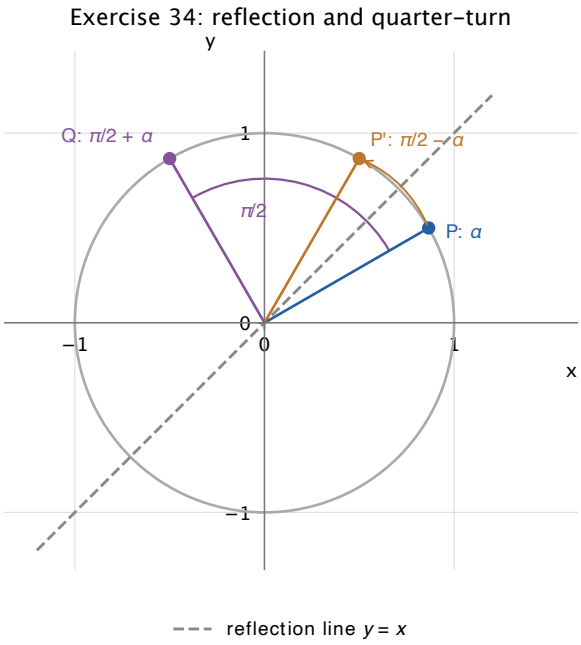



Figure 4: Unit-circle point P at angle alpha, its reflection P prime at pi over 2 minus alpha across y equals x, and its quarter-turn image Q at pi over 2 plus alpha

**Exercise 35.** Find the largest and smallest values of $2\sin^2\theta + 3\cos\theta$, and the values of $\cos\theta$ where they occur. (Hint: write the expression in terms of $c = \cos\theta$, with $-1 \le c \le 1$.)

**Solution.** Set $c = \cos\theta$, so $-1 \le c \le 1$ and $\sin^2\theta = 1 - c^2$. Then

$$2\sin^2\theta + 3\cos\theta = 2(1 - c^2) + 3c = -2\left(c - \frac{3}{4}\right)^2 + \frac{25}{8}.$$

The squared term is nonnegative, so the largest value is

$$\boxed{\frac{25}{8} \quad \text{when} \quad \cos\theta = \frac{3}{4}}.$$

This value of cosine is feasible because $3/4 \in [-1, 1]$.

To prove the minimum over the whole interval, subtract the candidate minimum $-3$ and factor:

$$(-2c^2 + 3c + 2) - (-3) = -2c^2 + 3c + 5 = (c + 1)(5 - 2c).$$

For $-1 \le c \le 1$, $c + 1 \ge 0$ and $5 - 2c \ge 3 > 0$. Thus the expression is at least $-3$, with equality exactly when $c = -1$. Consequently,

$$\boxed{\text{smallest value } = -3 \quad \text{when } \cos\theta = -1}.$$

As a check, the endpoint values are $-3$ at $c = -1$ and $3$ at $c = 1$, and the vertex value $25/8$ is larger than both.

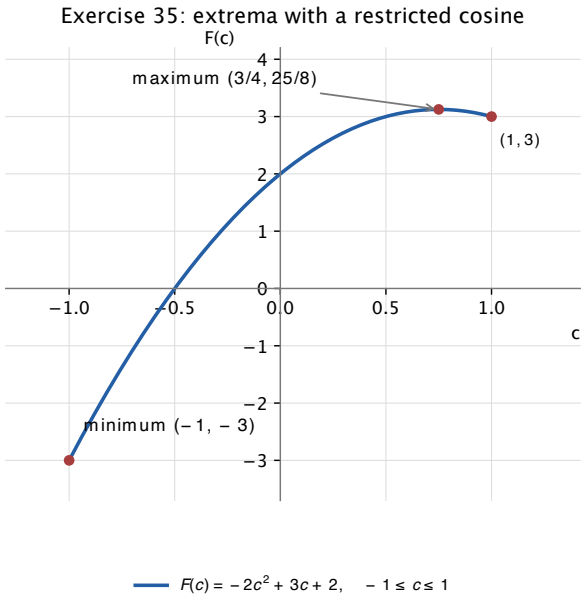



Figure 5: Quadratic F of c equals minus 2c squared plus 3c plus 2 on minus 1 to 1, with minimum minus 3 at c equals minus 1 and maximum 25 over 8 at c equals 3 over 4